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DC Motor High Current Diagnostic Log Sheet

Baseline log and test procedure for PMDC and geared motors drawing more current than expected

Start by recording correct baselines; a current is only high relative to a reference. Then use the decoupled no-load test to decide whether the extra current belongs to the motor or to the load before dismantling anything. Isolate and lock out the supply before uncoupling the motor, testing insulation or handling brush gear.

Step 1 — Measure current correctly

- [ ] Use a Hall-effect clamp meter with a true DC range; zero it with the jaws closed before clamping.
- [ ] Clamp one conductor only, on the motor side of the controller.
- [ ] On PWM drives record both the DC (average) reading and the true-RMS AC+DC reading; average sets torque, RMS sets heating.
- [ ] Battery-side current is roughly motor current × duty cycle, so it under-reads motor current at low speed.
- [ ] Record terminal voltage and speed at the same moment as current.

Step 2 — Establish baselines

- [ ] Full-load current from the nameplate.
- [ ] No-load current and no-load speed from the test sheet or commissioning record.
- [ ] Locked-rotor current calculated as (V − brush drop) ÷ armature resistance; do not measure it at full voltage.
- [ ] Starting current with a peak-hold clamp or oscilloscope over a normal start.
- [ ] Starts and reversals per hour from the machine cycle.

Step 3 — Decoupled no-load test

- [ ] Uncouple the motor and run it at rated voltage for a few minutes; record current and speed.
- [ ] No-load current within about 10 % of the baseline: the motor is healthy, look at the load side.
- [ ] Higher no-load current, normal speed: bearing or brush friction, or shorted turns.
- [ ] No-load speed above the original: weakened magnets; calculate Ke = (V − I0 × R − brush drop) ÷ ω0.
- [ ] Split the running current: load torque = Kt × (running current − no-load current).

Step 4 — Load-side checks

- [ ] Turn the driveline by hand with the motor removed; feel for tight spots and binding.
- [ ] Confirm the gear ratio fitted matches the drawing; motor torque = output torque ÷ (ratio × gear efficiency).
- [ ] Check chain and belt tension, coupling alignment, idlers and rollers.
- [ ] Measure brake coil voltage when released and check the air gap.

Step 5 — Motor electrical tests

- [ ] Insulation resistance at 500 V DC with controllers, encoders and suppression parts disconnected; investigate below about 1 MΩ.
- [ ] Bar-to-bar resistance around the commutator; readings should be uniform, typically within about 5 % of the average.
- [ ] Brush length, spring force and commutator condition.

- [ ] Oscilloscope check of PWM frequency and current ripple if RMS current is well above average current.

Step 6 — Protection coordination

- [ ] Controller current limit above normal acceleration current and within the motor's rated peak.
- [ ] Thermal protection (thermostat, thermistor or I^2t model) for sustained overload below the current limit.
- [ ] Fuse or breaker rated for DC at the system voltage, able to ride through starting current.
- [ ] Fuse or breaker rating not above the cable's current rating.

Baseline log (worked example, 24 V, 7.5 A, 2,550 rpm PMDC motor)

Measurement	Baseline	Measured	Interpretation
Terminal voltage	24.0 V	24.0 V	Supply correct
No-load current (decoupled)	0.60 A	0.90 A	+0.3 A internal friction: check bearings, brush springs
No-load speed (decoupled)	2,898 rpm	2,890 rpm	Ke 0.0748 vs 0.075: magnets healthy
Running current	7.5 A	9.8 A	+2.3 A above rated
Load torque Kt × (I − I0)	0.518 Nm	0.668 Nm	+29 %: driven machine binding
Copper loss ratio (I ÷ I rated)²	1.00	1.71	Motor running hot as a result
Locked-rotor current (calculated)	57.5 A	—	About 7.7 × rated: limit it with the controller

Values are an illustration for a fractional-kW PMDC motor. Always use your own motor's nameplate and test data, and follow the controller manufacturer's settings guidance.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.